

Project Title: Project 1: Password Strength Checker

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Objective To develop a defensive cybersecurity utility that evaluates password strength using strict length verification and pattern recognition. The tool acts as a logic skeleton to filter out weak credentials before encryption phases.

Methodology & Implementation The project was implemented in Python 3, leveraging standard string handling and conditional logic. To adhere to the requirement of linear $O(n)$ execution time, manual verbose loops were avoided. Instead, Python's C-optimized built-in `any()` function was utilized alongside generator expressions for short-circuit execution.

System Logic

1. **Length Verification:** Any input under 8 characters triggers an immediate fail to mitigate exponential brute force risks.
2. **Entropy Check:** The program evaluates the presence of uppercase characters, numerical digits, and symbols using the expanded Unicode entropy space.
3. **Classification:** Passwords are categorized as Weak, Medium, or Strong based on the combination of characters and overall length.

Testing & Results The script was verified against multiple input strings. Short passwords successfully triggered the immediate failure condition. Passwords utilizing full pattern recognition (uppercase, digits, symbols) and exceeding 12 characters were accurately flagged as strong.

Future Improvements Future iterations could integrate `hmac.compare_digest()` for constant-time comparisons if checking against known compromised password hashes, mitigating potential timing attacks as noted in advanced vulnerability research.

```

1  import string
2  def check_password_strength(password: str) -> str:
3      """
4      Evaluates the strength of a password based on length,
5      uppercase letters, digits, and special characters.
6      Runs with O(n) complexity using Pythonic short-circuiting.
7      """
8      if len(password) < 8:
9          return "Weak (Immediate Fail: Less than 8 characters)"
10     has_upper = any(char.isupper() for char in password)
11     has_digit = any(char.isdigit() for char in password)
12     has_symbol = any(char in string.punctuation for char in password)
13     score = 0
14     if has_upper:
15         score += 1
16     if has_digit:
17         score += 1
18     if has_symbol:
19         score += 1
20     if score == 3 and len(password) >= 12:
21         return "Strong"
22     elif score >= 2:
23         return "Medium"
24     else:
25         return "Weak (Lacks variety in character types)"
26 def main():
27     print("=== DecodeLabs Password Strength Checker ===")
28     print("Type 'exit' to quit the program.\n")
29     while True:
30         user_input = input("Enter a password to evaluate: ")
31         if user_input.lower() == 'exit':
32             print("Exiting program. Stay secure!")
33             break
34         strength = check_password_strength(user_input)
35         print(f"Result: {strength}\n")
36 if __name__ == "__main__":
37     main()

```

```
=== DecodeLabs Password Strength Checker ===  
Type 'exit' to quit the program.
```

```
Enter a password to evaluate: danial  
Result: Weak (Immediate Fail: Less than 8 characters)
```

```
Enter a password to evaluate: danial@!  
Result: Weak (Lacks variety in character types)
```

```
Enter a password to evaluate: Danial@1yhsk3#@$aewt  
Result: Strong
```

```
Enter a password to evaluate: exit  
Exiting program. Stay secure!
```

```
PS C:\Users\DELL\Desktop\password-strength-checker> █
```